

Geodesic Newton Direction with Blended RHS

1 Problem and notation

We solve the conic program

$$\min_x c^T x \quad \text{s.t.} \quad Ax + b \succeq 0,$$

where A is the constraint operator, b the affine term, c the cost, and $\succeq 0$ denotes membership in a symmetric cone with identity element e and quadratic representation $P(X)v$.

1.1 Perturbed system (θ -centering)

We consider a τ -scaled and θ -perturbed central path conditions:

$$A^T \lambda = \tau c + \theta (A^T e - c), \tag{1}$$

$$s = Ax + \tau b + \theta(e - b), \tag{2}$$

1.2 Geodesic parameterization

We use the Geodesic Newton direction d defined via

$$s = \frac{1}{k} P(w^{-1/2})(e - d), \quad \lambda = \frac{1}{k} P(w^{1/2})(e + d),$$

where $k = 1/\sqrt{\mu}$ and d is the Newton direction in v -space. The Gram matrix is $G = A^T P(w) A$.

2 Three-term decomposition

The Newton step at the current scaling W is decomposed into three components that are independent of any blending parameters.

2.1 Gram solves

$$G y_0 = A^T(2w),$$

$$G y_1^{(0)} = -(c + A^T P(w) b),$$

$$G y_1^{(\theta)} = c + A^T P(w) b - A^T (e + P(w) e).$$

2.2 Direction components

$$\begin{aligned} d_0 &= e - P(w^{1/2})(A y_0), \\ d_1^{(0)} &= -P(w^{1/2})(b + A y_1^{(0)}), \\ d_1^{(\theta)} &= P(w^{1/2})(b - e - A y_1^{(\theta)}). \end{aligned}$$

All six quantities $(y_0, y_1^{(0)}, y_1^{(\theta)}, d_0, d_1^{(0)}, d_1^{(\theta)})$ depend only on W , the original data (b, c) , and the constraint operator A . They are computed with one factorization and three back-solves.

3 Parameterized direction

Given independent parameters τ (original-problem weight) and θ (identity-centering weight), the Newton direction is

$$d(k, \tau, \theta) = d_0 + k(\tau d_1^{(0)} + \theta d_1^{(\theta)}),$$

which is affine in k , τ , and θ .

The corresponding variable is

$$y(k, \tau, \theta) = y_0 + k(\tau y_1^{(0)} + \theta y_1^{(\theta)}).$$

3.1 Special cases

- $\tau = 1, \theta = 0$: original problem, $d = d_0 + k d_1^{(0)}$.
- $\tau = 0, \theta = 1$: feasibility/centering problem with $b = e, c = A^T e$, $d = d_0 + k d_1^{(\theta)}$.
- $\tau = 1 - \theta$: single-parameter blend $b_\theta = \theta e + (1 - \theta) b, c_\theta = \theta A^T e + (1 - \theta) c$.

4 Minimum-norm k at fixed (τ, θ)

Define $\tilde{d}_1 = \tau d_1^{(0)} + \theta d_1^{(\theta)}$. The squared norm of the direction is

$$\|d(k)\|^2 = \|d_0\|^2 + 2k \langle d_0, \tilde{d}_1 \rangle + k^2 \|\tilde{d}_1\|^2,$$

which is minimized at

$$k^*(\tau, \theta) = -\frac{\langle d_0, \tau d_1^{(0)} + \theta d_1^{(\theta)} \rangle}{\|\tau d_1^{(0)} + \theta d_1^{(\theta)}\|^2}.$$

5 Duality identity

5.1 General form

For the QP with equality constraints

$$\min_x c^T x + \frac{1}{2} x^T Q x \quad \text{s.t.} \quad Ax + b \succeq 0, \quad Cx = d,$$

the KKT stationarity condition is $c + Qx = A^T \lambda + C^T \nu$. Multiplying by x :

$$c^T x + x^T Q x = \lambda^T (Ax) + \nu^T (Cx) = \langle s - b, \lambda \rangle + d^T \nu = \langle s, \lambda \rangle - b^T \lambda + d^T \nu.$$

In the HSD embedding with lifted variables ($x_{\text{phys}} = x/\tau$), the physical quadratic term is $(1/2)(x/\tau)^T Q (x/\tau) = x^T Q x / (2\tau^2)$. Combined with $\kappa = \mu/\tau$ and the central-path target $\langle s, \lambda \rangle + \kappa = \theta R$, the identity becomes

$$\boxed{b^T \lambda + c^T x + d^T \nu + \frac{x^T Q x}{2\tau} + \frac{\mu}{\tau} = \theta (\langle b, e \rangle + 1).}$$

At the trivially centered point ($\theta = 1$, $\tau = 1$, $k = 1$, $W = e$, $d = 0$): $\lambda = e$, $x = 0$, $\mu = 1$, and

$$b^T e + 0 + 0 + 0 + 1 = \langle b, e \rangle + 1. \quad \checkmark$$

5.2 Decomposition of the quadratic term

The primal variable $x(k, \tau, \theta) = y_0/k + \tau y_1^{(0)} + \theta y_1^{(\theta)}$ separates into a τ -free part and a τ -linear part. Define

$$f = f(k, \theta) = y_0/k + \theta y_1^{(\theta)}, \quad x = f + \tau y_1^{(0)}.$$

Then

$$x^T Q x = f^T Q f + 2\tau f^T Q y_1^{(0)} + \tau^2 (y_1^{(0)})^T Q y_1^{(0)},$$

and the $1/(2\tau)$ scaling gives three terms at distinct powers of τ :

$$\frac{x^T Q x}{2\tau} = \underbrace{\frac{f^T Q f}{2\tau}}_{\tau^{-1}} + \underbrace{f^T Q y_1^{(0)}}_{\tau^0} + \underbrace{\frac{\tau}{2} (y_1^{(0)})^T Q y_1^{(0)}}_{\tau^1}.$$

5.3 Duality residual

The absolute error in the identity is

$$r_{\text{dual}} = \left| b^T \lambda + c^T x + d^T \nu + \frac{x^T Q x}{2\tau} + \frac{\mu}{\tau} - \theta (\langle b, e \rangle + 1) \right|.$$

6 Joint (k, τ) selection with $\theta = \mu = 1/k^2$

Setting $\theta = \mu = 1/k^2$ ties the centering weight to the barrier parameter. The direction becomes

$$d = d_0 + k\tau d_1^{(0)} + \frac{1}{k} d_1^{(\theta)},$$

which depends on two free parameters (k, τ) .

6.1 Inner-product basis

All norms and inner products of d can be expressed in terms of six scalars computed once from the decomposition:

$$\begin{aligned} a &= \|d_0\|^2, & f &= \langle d_0, d_1^{(0)} \rangle, & g &= \langle d_0, d_1^{(\theta)} \rangle, \\ p &= \|d_1^{(0)}\|^2, & q &= \langle d_1^{(0)}, d_1^{(\theta)} \rangle, & r &= \|d_1^{(\theta)}\|^2. \end{aligned}$$

6.2 Optimal τ at fixed k

The squared norm

$$\|d\|^2 = a + 2k\tau f + \frac{2g}{k} + k^2\tau^2 p + 2\tau q + \frac{r}{k^2}$$

is quadratic in τ . Setting $\partial\|d\|^2/\partial\tau = 0$:

$$\tau^*(k) = -\frac{k f + q}{k^2 p}.$$

6.3 Reduced objective in k alone

Substituting $\tau^*(k)$ back:

$$\|d\|^2(k, \tau^*(k)) = a + \frac{2g}{k} + \frac{r}{k^2} - \frac{(k f + q)^2}{k^2 p}.$$

This is a rational function of k involving only the six precomputed scalars. It is minimized by a one-dimensional search over $k > 0$.

6.4 Weighted τ selection with duality constraint

The min-norm $\tau^*(k)$ above ignores the duality residual. To enforce the identity from Section 6, we select τ to minimize a weighted objective that penalizes both $\|d\|^2$ and the squared violation.

For fixed (k, θ) , expand the duality identity. The cone and linear-cost contributions are affine in τ ; the quadratic cost adds a τ^{-1} , τ^0 , and τ^1 term (Section 6.2):

$$\underbrace{(\sigma_0 + \gamma_0 + f^T Q y_1^{(0)})}_{\alpha} + \underbrace{(\sigma_1 + \gamma_1 + \frac{1}{2}(y_1^{(0)})^T Q y_1^{(0)})}_{\beta} \tau + \frac{\mu + \frac{1}{2}f^T Q f}{\tau} = \underbrace{\theta(\langle b, e \rangle + 1)}_R,$$

where

$$\begin{aligned}\sigma_0 &= \frac{1}{k} \langle b, P(w^{1/2})(e + d_0 + k\theta d_1^{(\theta)}) \rangle, & \sigma_1 &= \langle b, P(w^{1/2})(d_1^{(0)}) \rangle, \\ \gamma_0 &= c^T y_0/k + \theta c^T y_1^{(\theta)}, & \gamma_1 &= c^T y_1^{(0)},\end{aligned}$$

and $f = y_0/k + \theta y_1^{(\theta)}$ is the τ -free part of x . Multiplying through by τ gives the *violation quadratic*:

$$V(\tau) = \beta \tau^2 + (\alpha - R) \tau + \mu_{\text{eff}},$$

with

$$\begin{aligned}\beta &= \sigma_1 + \gamma_1 + \frac{1}{2} (y_1^{(0)})^T Q y_1^{(0)}, \\ \alpha &= \sigma_0 + \gamma_0 + f^T Q y_1^{(0)}, \\ \mu_{\text{eff}} &= \mu + \frac{1}{2} f^T Q f.\end{aligned}$$

When $Q = 0$ these reduce to the LP case. The three Q inner products $(y_1^{(0)})^T Q y_1^{(0)}$, $f^T Q y_1^{(0)}$, $f^T Q f$ are computed via `AccumulateQx`; only the first is τ -independent.

The $\|d\|^2$ terms, grouping the τ -independent part as $h = d_0 + k\theta d_1^{(\theta)}$, give

$$\|d\|^2 = \|h\|^2 + 2k \langle h, d_1^{(0)} \rangle \tau + k^2 p \tau^2.$$

The weighted objective is

$$J(\tau) = \|d\|^2 + w V(\tau)^2,$$

where $w > 0$ is a penalty weight. For $w \rightarrow \infty$ this enforces $V(\tau) = 0$ exactly (the duality identity); for $w = 0$ it recovers the pure min-norm selection.

The derivative $J'(\tau)$ is cubic in τ , so the minimizer can be found by golden-section search over $\tau > 0$.

6.5 Hard constraint ($w \rightarrow \infty$)

In the limit $w \rightarrow \infty$, τ is determined by the roots of the violation quadratic $V(\tau) = 0$:

$$\tau = \frac{(R - \alpha) \pm \sqrt{(R - \alpha)^2 - 4\beta \mu_{\text{eff}}}}{2\beta}.$$

When both roots are positive, choose the one with smaller $\|d\|^2$. When only one root is positive, use it. This enforces the duality identity exactly and eliminates the tuning parameter w .

6.6 Algorithm (per outer iteration)

1. Compute the decomposition $(d_0, d_1^{(0)}, d_1^{(\theta)})$: one factorization and three back-solves.
2. Compute the six inner products a, f, g, p, q, r .
3. Compute the duality coefficients σ_1, γ_1 (once per factorization) and σ_0, γ_0 (depend on k via θ).
4. For the current k (with $\theta = 1/k^2$), minimize $J(\tau)$ over $\tau > 0$.
5. Evaluate $d = d_0 + k\tau d_1^{(0)} + (1/k) d_1^{(\theta)}$.
6. Take the geodesic step $W \leftarrow W \cdot \exp(\alpha d)$ with $\alpha = \min(1, 2/\|d\|_\infty^2)$.

Centering terminates when both $\|d\|_\infty$ and the duality residual $|V(\tau)/\tau|$ fall below a threshold. The outer loop decreases $\theta = \mu = 1/k^2$ (e.g. $\theta \leftarrow 0.9\theta$) and re-centers at each level until $\theta < \varepsilon$.